

***Schmidtea mediterranea* hypoxia time-course bulk ATAC-seq analysis workflow**

Trimming of ATAC-seq reads

```
for sample in 0hR1 0hR2 0hR3 3hR1 3hR2 3hR3 12hR1 12hR2 12hR3 24hR1 24hR2
24hR3 36hR1 36hR2 36hR3
do
  cutadapt \
    -a CTGTCTCTTATACACATCT \
    -A CTGTCTCTTATACACATCT \
    -q 20,20 \
    -m 20 \
    -o ${sample}_1_trimmed.fq \
    -p ${sample}_2_trimmed.fq \
    ${sample}_1.fastq.gz ${sample}_2.fastq.gz
done
```

Alignment of ATAC-seq reads to Smed_rink.fa

```
for sample in 0hR1 0hR2 0hR3 3hR1 3hR2 3hR3 12hR1 12hR2 12hR3 24hR1 24hR2
24hR3 36hR1 36hR2 36hR3 do bowtie2 --very-sensitive -X 2000 -p 8 -x Smed_Rink \ -1
${sample}_1_trimmed.fq \ -2 ${sample}_2_trimmed.fq \ | samtools view -bS - \ |
samtools sort -o ${sample}.sorted.bam - && samtools index ${sample}.sorted.bam
done
```

Obtaining alignment summary of all ATAC-seq samples

```
(ATAC-seq) lady7197@Bio-L-SV-
002001:/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment$ cat > atac_alignment_summary.py <<'PY'
> #!/usr/bin/env python3
>
>
> from pathlib import Path
> import csv
> import re
> import subprocess
> import sys
>
>
> def require_tool(tool: str) -> None:
>     from shutil import which
>     if which(tool) is None:
>         print(f"Error: '{tool}' is not on PATH.", file=sys.stderr)
>         sys.exit(1)
>
>
> def run_cmd(cmd: list[str]) -> str:
```

```

> try:
>     result = subprocess.run(cmd, check=True, capture_output=True, text=True)
>     return result.stdout
> except subprocess.CalledProcessError as e:
>     print(f"Error running: {' '.join(cmd)}", file=sys.stderr)
>     if e.stderr:
>         print(e.stderr, file=sys.stderr)
>     sys.exit(1)
>
>
> def find_int(pattern: str, text: str) -> int:
>     m = re.search(pattern, text, flags=re.MULTILINE)
>     return int(m.group(1)) if m else 0
>
>
> def find_pair_int(pattern: str, text: str) -> tuple[int, int]:
>     m = re.search(pattern, text, flags=re.MULTILINE)
>     if not m:
>         return 0, 0
>     return int(m.group(1)), int(m.group(2))
>
>
> def parse_flagstat(flagstat_text: str) -> dict[str, int]:
>     stats = {}
>     stats["total_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+in total\b", flagstat_text)
>     stats["secondary_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+secondary\b",
flagstat_text)
>     stats["supplementary_reads"] =
find_int(r"^\s*(\d+)\s+\+\s+\d+\s+supplementary\b", flagstat_text)
>     stats["duplicates_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+duplicates\b",
flagstat_text)
>     stats["mapped_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+mapped\b",
flagstat_text)
>     stats["paired_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+paired in sequencing\b",
flagstat_text)
>     stats["read1_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+read1\b", flagstat_text)
>     stats["read2_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+read2\b", flagstat_text)
>     stats["properly_paired_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+properly
paired\b", flagstat_text)
>     stats["with_itself_mate_mapped"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+with itself
and mate mapped\b", flagstat_text)
>     stats["singletons_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+singletons\b",
flagstat_text)
>     stats["mate_diff_chr_mapq5_reads"] = find_int(r"^\s*(\d+)\s+\+\s+\d+\s+with mate
mapped to a different chr \(\mapQ>=5\)\b", flagstat_text)
>     return stats
>

```

```

>
> def calc_pct(num: int, denom: int) -> float:
>     return (100.0 * num / denom) if denom else 0.0
>
>
> def main() -> None:
>     require_tool("samtools")
>
>     bam_files = sorted(Path(".").glob("*.sorted.bam"))
>     if not bam_files:
>         print("No *.sorted.bam files found in the current directory.", file=sys.stderr)
>         sys.exit(1)
>
>     out_path = Path("atac_alignment_summary.tsv")
>
>     header = [
>         "sample",
>         "bam_file",
>         "total_reads",
>         "mapped_reads",
>         "mapped_pct",
>         "paired_reads",
>         "properly_paired_reads",
>         "properly_paired_pct",
>         "duplicates_reads",
>         "duplicates_pct",
>         "singletons_reads",
>         "singletons_pct",
>         "secondary_reads",
>         "supplementary_reads",
>         "read1_reads",
>         "read2_reads",
>         "with_itself_mate_mapped_reads",
>         "mate_diff_chr_mapq5_reads",
>         "chrM_mapped_reads",
>         "chrM_pct_of_mapped",
>     ]
>
>     with out_path.open("w", newline="") as out_f:
>         writer = csv.writer(out_f, delimiter="\t")
>         writer.writerow(header)
>         out_f.flush()
>
>         for i, bam in enumerate(bam_files, start=1):
>             sample = bam.name.replace(".sorted.bam", "")
>
>             flagstat = run_cmd(["samtools", "flagstat", str(bam)])

```

```

> fs = parse_flagstat(flagstat)
>
> idxstats = run_cmd(["samtools", "idxstats", str(bam)])
> chrM_mapped = 0
> for line in idxstats.strip().splitlines():
>     fields = line.split("\t")
>     if len(fields) >= 3 and fields[0] in {"MT", "chrM", "M",
"mitochondrion_genome"}:
>         try:
>             chrM_mapped += int(fields[2])
>         except ValueError:
>             pass
>
> mapped_pct = calc_pct(fs["mapped_reads"], fs["total_reads"])
> properly_paired_pct = calc_pct(fs["properly_paired_reads"], fs["total_reads"])
> duplicates_pct = calc_pct(fs["duplicates_reads"], fs["total_reads"])
> singletons_pct = calc_pct(fs["singletons_reads"], fs["total_reads"])
> chrM_pct_of_mapped = calc_pct(chrM_mapped, fs["mapped_reads"])
>
> writer.writerow([
>     sample,
>     bam.name,
>     fs["total_reads"],
>     fs["mapped_reads"],
>     f'{mapped_pct:.2f}',
>     fs["paired_reads"],
>     fs["properly_paired_reads"],
>     f'{properly_paired_pct:.2f}',
>     fs["duplicates_reads"],
>     f'{duplicates_pct:.2f}',
>     fs["singletons_reads"],
>     f'{singletons_pct:.2f}',
>     fs["secondary_reads"],
>     fs["supplementary_reads"],
>     fs["read1_reads"],
>     fs["read2_reads"],
>     fs["with_itself_mate_mapped"],
>     fs["mate_diff_chr_mapq5_reads"],
>     chrM_mapped,
>     f'{chrM_pct_of_mapped:.2f}',
> ])
> out_f.flush()
>
> print(f"[{i}/{len(bam_files)}] finished {sample}", flush=True)
>
> print(f"Done. Wrote {out_path.resolve()}")
>

```

```

>
> if __name__ == "__main__":
>     main()
> PY
(ATAC-seq) lady7197@Bio-L-SV-
002001:/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment$
(ATAC-seq) lady7197@Bio-L-SV-
002001:/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment$ python3 -u atac_alignment_summary.py
[1/15] finished 0hR1
[2/15] finished 0hR2
[3/15] finished 0hR3
[4/15] finished 12hR1
[5/15] finished 12hR2
[6/15] finished 12hR3
[7/15] finished 24hR1
[8/15] finished 24hR2
[9/15] finished 24hR3
[10/15] finished 36hR1
[11/15] finished 36hR2
[12/15] finished 36hR3
[13/15] finished 3hR1
[14/15] finished 3hR2
[15/15] finished 3hR3
Done. Wrote /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment/atac_alignment_summary.tsv

```

Conducting ATAC filtering

```

for bam in *.sorted.bam; do
    sample=${bam%.sorted.bam}

    echo "Processing $sample"

    samtools view -b -f 2 -q 30 "$bam" > "${sample}.filtered.bam"
    samtools sort -o "${sample}.filtered.sorted.bam" "${sample}.filtered.bam"
    samtools index "${sample}.filtered.sorted.bam"

    rm "${sample}.filtered.bam"
done

```

Filtering out mitochondrial reads

Smed_Rink.fa already excludes mitochondrial contigs (Pérez-Posada *et al.*, 2025).

Calculating unique mapping rate

```
cd /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis/alignment
```

```
echo -e  
"Sample\tProperlyPairedReads\tUniqueMappedReads_MAPQ30\tUniqueMappingRate  
Pct" \  
> unique_mapping_rate.tsv
```

```
for bam in *.sorted.bam  
do
```

```
    sample=${bam%.sorted.bam}
```

```
#####  
#####
```

```
    # ALL PROPERLY PAIRED READS
```

```
#####  
#####
```

```
    proper=$(samtools view -@ 8 -c -f 2 "$bam")
```

```
#####  
#####
```

```
    # UNIQUE PROPERLY PAIRED READS
```

```
    # MAPQ >=30
```

```
#####  
#####
```

```
    unique=$(samtools view -@ 8 -c -f 2 -q 30 "$bam")
```

```
#####  
#####
```

```
    # UNIQUE MAPPING RATE
```

```
#####  
#####
```

```
    rate=$(awk -v u="$unique" -v p="$proper" \  
    'BEGIN{printf "%.2f", (u/p)*100}')
```

```
#####  
#####
```

```
# OUTPUT
```

```
#####  
#####
```

```
echo -e "${sample}\t${proper}\t${unique}\t${rate}" \  
| tee -a unique_mapping_rate.tsv
```

```
done
```

Conducting de-duplication

```
mkdir -p dedup_bam_filtered tmp_fixmate
```

```
for bam in *.filtered.sorted.bam; do  
[ -e "$bam" ] || continue
```

```
sample=${bam%.filtered.sorted.bam}  
echo "Processing $sample at $(date)"
```

```
samtools sort -@ 16 -n -o tmp_fixmate/${sample}.name_sorted.bam "$bam"  
samtools fixmate -@ 16 -m tmp_fixmate/${sample}.name_sorted.bam  
tmp_fixmate/${sample}.fixmate.bam  
samtools sort -@ 16 -o tmp_fixmate/${sample}.positionsorted.bam  
tmp_fixmate/${sample}.fixmate.bam  
samtools markdup -@ 16 -r tmp_fixmate/${sample}.positionsorted.bam  
dedup_bam_filtered/${sample}.dedup.bam  
samtools index dedup_bam_filtered/${sample}.dedup.bam
```

```
rm -f tmp_fixmate/${sample}.name_sorted.bam \  
tmp_fixmate/${sample}.fixmate.bam \  
tmp_fixmate/${sample}.positionsorted.bam
```

```
echo "Finished $sample at $(date)"  
done
```

```
cd /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis/alignment
```

```
echo -e  
"Sample\tPreDedupReads\tPostDedupReads\tPercentRetained\tPercentDuplicatesRe  
moved"
```

```
for dedup in dedup_bam_filtered/*.dedup.bam  
do
```

```

sample=$(basename "$dedup" .dedup.bam)

original="${sample}.filtered.sorted.bam"

pre=$(samtools view -@ 8 -c "$original")

post=$(samtools view -@ 8 -c "$dedup")

retained=$(awk -v p="$post" -v q="$pre" 'BEGIN{printf "%.2f", (p/q)*100}')

removed=$(awk -v p="$post" -v q="$pre" 'BEGIN{printf "%.2f", 100-((p/q)*100)}')

echo -e "${sample}\t${pre}\t${post}\t${retained}\t${removed}"

done

cd /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment

echo -e
"Sample\tTotalReads\tFilteredReads\tDeduplicatedReads\tFilteringRetentionPct\tFinalRetentionPct"

for dedup in dedup_bam_filtered/*.dedup.bam
do

    sample=$(basename "$dedup" .dedup.bam)

    totalbam="${sample}.sorted.bam"

    filteredbam="${sample}.filtered.sorted.bam"

#####
#####
# COUNTS

#####
#####

    total=$(samtools view -@ 8 -c "$totalbam")

    filtered=$(samtools view -@ 8 -c "$filteredbam")

    deduped=$(samtools view -@ 8 -c "$dedup")

```



```
#####
#####
# PERCENTAGES

#####
#####

filter_pct=$(awk -v f="$filtered" -v t="$total" \
'BEGIN{printf "%.2f", (f/t)*100}')

final_pct=$(awk -v d="$deduped" -v t="$total" \
'BEGIN{printf "%.2f", (d/t)*100}')

#####
#####
# OUTPUT

#####
#####

echo -e "${sample}\t$total\t$filtered\t$deduped\t$filter_pct\t$final_pct"

done \
| tee ATAC_final_mapping_summary.tsv

cd /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment
```

Calculating fragment size distribution

```
cd /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment

mkdir -p fragment_size_distribution/plots

for bam in dedup_bam/*.dedup.bam; do
    sample=$(basename "$bam" .dedup.bam)

    samtools view -f 66 -F 3852 "$bam" \
    | awk '{
        t = $9
        if (t < 0) t = -t
        if (t > 0 && t < 1000) print t
    }' > fragment_size_distribution/${sample}.fragment_lengths.txt
done
```

```

cat > plot_fragments.py << 'EOF'
import os
import glob
import matplotlib.pyplot as plt

outdir = "fragment_size_distribution"
plotdir = os.path.join(outdir, "plots")
os.makedirs(plotdir, exist_ok=True)

files = sorted(glob.glob(os.path.join(outdir, "*.fragment_lengths.txt")))
all_data = {}

for f in files:
    sample = os.path.basename(f).replace(".fragment_lengths.txt", "")
    vals = []

    with open(f) as fh:
        for line in fh:
            try:
                x = int(line.strip())
                if 0 < x < 1000:
                    vals.append(x)
            except ValueError:
                pass

    all_data[sample] = vals

    plt.figure(figsize=(8, 5))
    plt.hist(vals, bins=200, range=(0, 800))
    plt.xlabel("Fragment length (bp)")
    plt.ylabel("Count")
    plt.title(sample)
    plt.tight_layout()
    plt.savefig(os.path.join(plotdir, f"{sample}.png"), dpi=200)
    plt.close()

plt.figure(figsize=(10, 7))

for sample, vals in all_data.items():
    plt.hist(
        vals,
        bins=200,
        range=(0, 800),
        histtype="step",
        linewidth=1.5,
        density=True,

```

```

        label=sample,
    )

plt.xlabel("Fragment length (bp)")
plt.ylabel("Density")
plt.title("Fragment size distribution across all samples")
plt.legend(fontsize=7, ncol=2)
plt.tight_layout()
plt.savefig(os.path.join(plotdir, "combined_fragment_size_distribution.png"), dpi=200)
plt.close()
EOF

```

```
python3 plot_fragments.py
```

```

ls -lh fragment_size_distribution
ls -lh fragment_size_distribution/plots

```

Plotting combined fragment size distribution

```
/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment
```

```

cat > make_fragment_style_plot.py <<'EOF'
import os
import re
import glob
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.cm as cm

outdir = "fragment_size_distribution"
plotdir = os.path.join(outdir, "plots")
os.makedirs(plotdir, exist_ok=True)

files = sorted(glob.glob(os.path.join(outdir, "*.fragment_lengths.txt")))
if not files:
    raise SystemExit("No fragment length files found")

def sort_key(path):
    sample = os.path.basename(path).replace(".fragment_lengths.txt", "")
    m = re.match(r"^(?P<id>\d+)hR(?P<len>\d+)$", sample)
    if m:
        return (int(m.group(1)), int(m.group(2)), sample)
    return (9999, 9999, sample)

files = sorted(files, key=sort_key)
samples = [os.path.basename(f).replace(".fragment_lengths.txt", "") for f in files]

```

```

bin_edges = np.arange(0, 501, 1)
x = np.arange(0.5, 500.0, 1.0)

all_hist = []

for f in files:
    vals = np.loadtxt(f, dtype=float)
    if vals.ndim == 0:
        vals = np.array([vals], dtype=float)

    vals = vals[(vals >= 0) & (vals <= 1000)]

    hist, _ = np.histogram(vals, bins=bin_edges, density=True)

    # start curves at 35 bp
    hist = hist.astype(float)
    hist[x < 35] = np.nan

    all_hist.append(hist)

# red-blue palette
colors = cm.get_cmap("RdBu")(np.linspace(0.05, 0.95, len(samples)))

plt.figure(figsize=(12, 12))
ax = plt.gca()

for sample, hist, color in zip(samples, all_hist, colors):
    ax.plot(x, hist, color=color, linewidth=3.2, alpha=0.9, label=sample)

ax.set_xlim(0, 500)
ax.set_xlabel("\nFragment length (bp)", fontsize=20)
ax.set_ylabel("Normalised read density\n", fontsize=20)

ax.tick_params(axis="x", labelsz=15)
ax.tick_params(axis="y", labelsz=15)

for xpos in [50, 150, 350]:
    ax.axvline(xpos, linestyle="dashed", color="black", linewidth=1)

leg = ax.legend(
    title="Sample",
    bbox_to_anchor=(1.02, 0.5),
    loc="center left",
    frameon=False,
    ncol=1,
    fontsize=14,
    title_fontsize=16

```

```
)

for line in leg.get_lines():
    line.set_linewidth(3.2)

plt.tight_layout(rect=[0, 0, 0.78, 1])
plt.savefig(os.path.join(plotdir, "combined_fragment_size_distribution.png"), dpi=300)

print("Done")
EOF
```

```
python3 make_fragment_style_plot.py
```

Plotting ATAC-seq sample-to-sample correlation heatmap

```
# Plotting ATAC-seq sample-to-sample correlation heatmap
```

```
conda activate ATAC-seq
```

```
cd /drives/raid/AboobakerLab/James/hypoxia_ATAC-
seq_timecourse/april_2026_analysis/alignment
```

```
mkdir -p atac_correlation
```

```
multiBigwigSummary bins \
-b bigwig/0hR1.bw \
  bigwig/0hR2.bw \
  bigwig/0hR3.bw \
  bigwig/3hR1.bw \
  bigwig/3hR2.bw \
  bigwig/3hR3.bw \
  bigwig/12hR1.bw \
  bigwig/12hR2.bw \
  bigwig/12hR3.bw \
  bigwig/24hR1.bw \
  bigwig/24hR2.bw \
  bigwig/24hR3.bw \
  bigwig/36hR1.bw \
  bigwig/36hR2.bw \
  bigwig/36hR3.bw \
--labels \
  0hR1 0hR2 0hR3 \
  3hR1 3hR2 3hR3 \
  12hR1 12hR2 12hR3 \
  24hR1 24hR2 24hR3 \
  36hR1 36hR2 36hR3 \
-o atac_correlation/ATAC_bigwig_summary.npz
```

```

plotCorrelation \
-in atac_correlation/ATAC_bigwig_summary.npz \
--corMethod spearman \
--whatToPlot heatmap \
--plotNumbers \
--skipZeros \
--labels \
  0hR1 0hR2 0hR3 \
  3hR1 3hR2 3hR3 \
  12hR1 12hR2 12hR3 \
  24hR1 24hR2 24hR3 \
  36hR1 36hR2 36hR3 \
-outFileCorMatrix atac_correlation/ATAC_correlation_matrix.tsv \
-o atac_correlation/ATAC_correlation_heatmap.pdf

```

```
ls -lh atac_correlation/ATAC_correlation_heatmap.pdf
```

Plotting ATAC-seq PCA plot

```
# Plotting ATAC-seq PCA
```

```
conda activate ATAC-seq
```

```
cd /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment
```

```
mkdir -p atac_pca
```

```

multiBigwigSummary bins \
-b bigwig/0hR1.bw \
  bigwig/0hR2.bw \
  bigwig/0hR3.bw \
  bigwig/3hR1.bw \
  bigwig/3hR2.bw \
  bigwig/3hR3.bw \
  bigwig/12hR1.bw \
  bigwig/12hR2.bw \
  bigwig/12hR3.bw \
  bigwig/24hR1.bw \
  bigwig/24hR2.bw \
  bigwig/24hR3.bw \
  bigwig/36hR1.bw \
  bigwig/36hR2.bw \
  bigwig/36hR3.bw \
--labels \
  0hR1 0hR2 0hR3 \

```

```
3hR1 3hR2 3hR3 \
12hR1 12hR2 12hR3 \
24hR1 24hR2 24hR3 \
36hR1 36hR2 36hR3 \
-o atac_pca/ATAC_summary.npz
```

```
plotPCA \
-in atac_pca/ATAC_summary.npz \
-o atac_pca/ATAC_PCA.pdf \
--labels \
0hR1 0hR2 0hR3 \
3hR1 3hR2 3hR3 \
12hR1 12hR2 12hR3 \
24hR1 24hR2 24hR3 \
36hR1 36hR2 36hR3
```

```
ls -lh atac_pca/ATAC_PCA.pdf
```

```
python - <<'PY'
import re
import numpy as np
import matplotlib.pyplot as plt
```

```
npz_file = "atac_pca/ATAC_summary.npz"
```

```
out_pdf = "atac_pca/ATAC_PCA_timepoint_colored.pdf"
out_png = "atac_pca/ATAC_PCA_timepoint_colored.png"
```

```
npz = np.load(npz_file, allow_pickle=True)
```

```
# Load matrix
```

```
X = None
```

```
for k in ("matrix", "readCounts", "counts"):
```

```
    if k in npz.files:
```

```
        X = np.asarray(npz[k], dtype=float)
```

```
        break
```

```
if X is None:
```

```
    raise KeyError(f"No matrix-like array found in {npz.files}")
```

```
# Load labels
```

```
labels = None
```

```
for k in ("labels", "sampleLabels"):
```

```
    if k in npz.files:
```

```
        labels = [str(x) for x in npz[k]]
```

```
        break
```

```

if labels is None:
    raise KeyError(f"No labels found in {npz.files}")

# Ensure samples are rows
if X.shape[0] != len(labels) and X.shape[1] == len(labels):
    X = X.T

if X.shape[0] != len(labels):
    raise ValueError(f"Shape mismatch: X={X.shape}, labels={len(labels)}")

# Remove constant bins
X = np.nan_to_num(X, nan=0.0, posinf=0.0, neginf=0.0)

keep = X.var(axis=0) > 0
X = X[:, keep]

# Z-score each genomic bin
mu = X.mean(axis=0)
sd = X.std(axis=0, ddof=1)
sd[sd == 0] = 1.0

Xz = (X - mu) / sd

# PCA via SVD
U, S, Vt = np.linalg.svd(Xz, full_matrices=False)

pcs = U[:, :2] * S[:2]

eigvals = (S**2) / (Xz.shape[0] - 1)
var_ratio = eigvals / eigvals.sum()

def get_timepoint(label):
    m = re.match(r"(\d+h)", label)
    return m.group(1) if m else label

timepoints = [get_timepoint(x) for x in labels]

# Timepoint colours
colors = {
    "0h": "#000000",
    "3h": "#009E73",
    "12h": "#56B4E9",
    "24h": "#D2B48C",
    "36h": "#7B3294",
}

order = ["0h", "3h", "12h", "24h", "36h"]

```



```

fig, ax = plt.subplots(figsize=(8.2, 6.2))

for tp in order:

    idx = [i for i, t in enumerate(timepoints) if t == tp]

    if not idx:
        continue

    ax.scatter(
        pcs[idx, 0],
        pcs[idx, 1],
        s=55,
        marker="o",
        c=colors[tp],
        edgecolors="none",
        label=tp,
        alpha=0.95,
    )

    ax.axhline(0, color="black", linestyle=":", linewidth=1)
    ax.axvline(0, color="black", linestyle=":", linewidth=1)

    ax.set_xlabel(
        f"PC1 ({var_ratio[0]*100:.1f}% of variance explained)"
    )

    ax.set_ylabel(
        f"PC2 ({var_ratio[1]*100:.1f}% of variance explained)"
    )

    ax.set_title("ATAC-seq PCA")

    ax.legend(
        title="Timepoint",
        frameon=True,
        loc="center left",
        bbox_to_anchor=(1.02, 0.5)
    )

plt.tight_layout()

plt.savefig(out_pdf)
plt.savefig(out_png, dpi=300)

print("Overwritten:")

```

```
print(out_pdf)
print(out_png)
PY
```

Calling ATAC-seq peaks using MACS3

```
# Calling ATAC-seq peaks using MACS3
```

```
cd /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment
```

```
conda activate macs3_env
```

```
INPUT=/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis/alignment/dedup_bam_filtered
```

```
mkdir -p peaks
mkdir -p logs
```

```
for bam in "$INPUT"/*.dedup.bam
do
```

```
    sample=$(basename "$bam" .dedup.bam)
```

```
    if [ ! -f "peaks/${sample}_peaks.narrowPeak" ]; then
```

```
        echo "Calling peaks for $sample"
```

```
        macs3 callpeak \
        -t "$bam" \
        -f BAMPE \
        -g 840213658 \
        -n "$sample" \
        --outdir peaks \
        --keep-dup 1 \
        -q 1e-4 \
        > "logs/${sample}.macs3.out" \
        2> "logs/${sample}.macs3.err"
```

```
    fi
```

```
done
```

```
ls peaks/*_peaks.narrowPeak | wc -l
```

```
wc -l peaks/*_peaks.narrowPeak
```

Calculating fraction of reads in peaks (FRiP) score

```
conda create -n frip_env -c bioconda -c conda-forge subread -y
conda activate frip_env
```

which featureCounts

```
BASE=/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis
```

```
rm -rf \
"${BASE}/FrIP/frip_saf" \
"${BASE}/FrIP/frip_counts" \
"${BASE}/FrIP/frip_scores.txt"
```

```
cat > "${BASE}/FrIP/run_frip.sh" <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
```

```
# FRiP calculation script
#
# FRiP = fraction of reads in peaks
# Calculated as:
# Assigned reads / total reads
# where totals are derived from featureCounts summary output
```

```
BASE=/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-
course/april_2026_analysis
INPUT=${BASE}/alignment/dedup_bam_filtered
PEAKS=${BASE}/alignment/tss_enrichment/peaks
OUTDIR=${BASE}/FrIP
```

```
mkdir -p "${OUTDIR}/frip_saf" "${OUTDIR}/frip_counts"
```

```
# Check featureCounts exists
command -v featureCounts >/dev/null 2>&1 || {
    echo "ERROR: featureCounts not found. Activate frip_env." >&2
    exit 1
}
```

```
tmp_scores="${OUTDIR}/frip_scores.tmp.tsv"
echo -e "Sample\tFRiP" > "$tmp_scores"
```

```
shopt -s nullglob
```

```
for peak in "${PEAKS}"/*_peaks.narrowPeak; do
```

```

sample=$(basename "$peak" _peaks.narrowPeak)
bam="${INPUT}/${sample}.dedup.bam"
saf="${OUTDIR}/frip_saf/${sample}.saf"
out="${OUTDIR}/frip_counts/${sample}.txt"

# Sanity check
if [[ ! -f "$bam" ]]; then
    echo "ERROR: Missing BAM file: $bam" >&2
    exit 1
fi

# Create SAF annotation from peaks
awk 'BEGIN{OFS="\t"; print "GeneID","Chr","Start","End","Strand"}
     {print "peak_"NR,$1,$2+1,$3,"."}' "$peak" > "$saf"

npeaks=$(wc -l < "$saf")
echo "Processing ${sample} (${npeaks} peaks)"

# Count reads in peaks
featureCounts \
-F SAF \
-a "$saf" \
-o "$out" \
-p \
--countReadPairs \
"$bam" >/dev/null

# Calculate FRiP
assigned=$(awk ' $1=="Assigned"{print $2}' "${out}.summary")
total=$(awk 'NR>1 {sum+=$2} END{print sum+0}' "${out}.summary")

if [[ -z "$assigned" || "$total" -eq 0 ]]; then
    echo "ERROR: Could not parse featureCounts summary for ${sample}" >&2
    exit 1
fi

frip=$(awk -v a="$assigned" -v t="$total" 'BEGIN{printf "%.4f", a/t}')

echo -e "${sample}\t${frip}" >> "$tmp_scores"

done

# Sort output (natural order)
{
    head -n 1 "$tmp_scores"
    tail -n +2 "$tmp_scores" | sort -V
} > "${OUTDIR}/frip_scores.txt"

```

```
rm -f "$tmp_scores"
```

```
echo "FRiP calculation complete:"  
echo "${OUTDIR}/frip_scores.txt"  
EOF
```

```
chmod +x "${BASE}/FrIP/run_frip.sh"
```

```
cd "${BASE}/alignment"  
conda activate frip_env
```

```
"${BASE}/FrIP/run_frip.sh"
```

```
cat "${BASE}/FrIP/frip_scores.txt"
```

Conducting DiffBind analysis

```
#!/usr/bin/env Rscript
```

```
suppressPackageStartupMessages({  
  library(DiffBind)  
  library(dplyr)  
})
```

```
# =====
```

```
# DiffBind consensus peakset generation
```

```
#
```

```
# Input:
```

```
# - diffbind_samplesheet.csv
```

```
# - alignment/dedup_bam_filtered/*.dedup.bam
```

```
# - alignment/tss_enrichment/peaks/*_peaks.narrowPeak
```

```
#
```

```
# Output:
```

```
# - diffbind_samplesheet_absolute.csv
```

```
# - consensus_peakset.csv
```

```
# - consensus_peakset.bed
```

```
#
```

```
# =====
```

```
# -----
```

```
# Paths
```

```
# -----
```

```
base_dir <- "/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis"
```

```
align_dir <- file.path(base_dir, "alignment")
```

```

input_sheet <- file.path(
  base_dir,
  "DiffBind",
  "diffbind_samplesheet.csv"
)

outdir <- file.path(
  base_dir,
  "DiffBind",
  "consensus_peakset"
)

dir.create(outdir, showWarnings = FALSE, recursive = TRUE)

abs_sheet <- file.path(
  outdir,
  "diffbind_samplesheet_absolute.csv"
)

# -----
# Building absolute sample sheet
# -----
message("Building absolute sample sheet...")

sheet <- read.csv(
  input_sheet,
  stringsAsFactors = FALSE
)

required_cols <- c(
  "SampleID",
  "Tissue",
  "Condition",
  "Replicate",
  "bamReads",
  "Peaks",
  "PeakCaller"
)

missing_cols <- setdiff(
  required_cols,
  names(sheet)
)

if(length(missing_cols) > 0) {
  stop(

```

```

    "Input sample sheet is missing columns: ",
    paste(missing_cols, collapse = ", ")
  )
}

# Absolute BAM paths
sheet$bamReads <- file.path(
  align_dir,
  "dedup_bam_filtered",
  paste0(sheet$SampleID, ".dedup.bam")
)

# Absolute peak paths
sheet$Peaks <- file.path(
  align_dir,
  "tss_enrichment/peaks",
  paste0(sheet$SampleID, "_peaks.narrowPeak")
)

write.csv(
  sheet,
  abs_sheet,
  row.names = FALSE
)

# -----
# Building DiffBind object
# -----
message("Building DiffBind object...")

db <- dba(
  sampleSheet = abs_sheet
)

# -----
# Creating consensus peakset
# -----
message("Generating consensus peakset...")

db <- dba.count(
  db,
  minOverlap = 2,
  summits = 250
)

# -----
# Extracting consensus peaks

```

```

# -----
message("Exporting consensus peakset...")

consensus_peaks <- dba.peakset(
  db,
  bRetrieve = TRUE
)

consensus_df <- as.data.frame(
  consensus_peaks
)

# CSV export
write.csv(
  consensus_df,
  file.path(
    outdir,
    "consensus_peakset.csv"
  ),
  row.names = FALSE
)

# BED export
bed_df <- consensus_df[, c("Chr", "Start", "End")]

write.table(
  bed_df,
  file.path(
    outdir,
    "consensus_peakset.bed"
  ),
  sep = "\t",
  quote = FALSE,
  row.names = FALSE,
  col.names = FALSE
)

message("Consensus peakset generation complete.")

```


Conducting TOBIAS analysis

```
(Tobias_env) lady7197@Bio-L-SV-002001:/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-course/april_2026_analysis/TOBIAS$
```

```
nano run_tobias_whitelist.sh
```

```
#!/usr/bin/env bash
set -euo pipefail
```

```
BASE=/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-course/april_2026_analysis
TOBIAS_DIR="$BASE/TOBIAS"
OUT="$BASE/TOBIAS/TOBIAS_hypoxia"
BAMDIR="$BASE/alignment/dedup_bam_filtered"
PEAKS="$BASE/hypoxia_ATAC_peaks.corrected.fixed.bed"
# (produced before by PEAKS=/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-course/april_2026_analysis/hypoxia_ATAC_peaks.corrected.bed
awk 'BEGIN{FS=OFS="\t"} NF>=3 {if ($2 < 0) $2 = 0; if ($3 > $2) print}' "$PEAKS" >
/drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-course/april_2026_analysis/hypoxia_ATAC_peaks.corrected.fixed.bed) #
```

```
GENOME="$BASE/Smed_Rink.fa"
JASPAR_ALL="$BASE/TOBIAS/JASPAR2026_CORE_non-redundant_pfms_jaspar.txt"
FILTERED_MOTIFS="$OUT/hypoxia_ATAC_peaks.animal_only_whitelist.jaspar"
```

```
rm -rf "$OUT"/{merged_bam,ATACCorrect,ScoreBigwig,BINDetect,plots,logs}
"$FILTERED_MOTIFS"
mkdir -p "$OUT"/{merged_bam,ATACCorrect,ScoreBigwig,BINDetect,plots,logs}
```

```
source "$(conda info --base)/etc/profile.d/conda.sh"
#####
#####
# Building a whitelist-filtered JASPAR file for TOBIAS BINDetect
#####
#####
```

```
python3 - "$JASPAR_ALL" "$METAZOAN_NON-REDUNDANT_MOTIFS" <<'PY'
import sys
```

```
jaspar_file = sys.argv[1]
filtered_out = sys.argv[2]
```

```
allowed_motif_ids = set([
```

"MA0010.1","MA0011.1","MA0012.1","MA0013.1","MA0015.1","MA0016.1","MA0022.1",
"MA0023.1","MA0026.1","MA0049.1",

"MA0085.1","MA0086.1","MA0094.1","MA0126.1","MA0165.1","MA0166.1","MA0167.1",
"MA0168.1","MA0169.1","MA0170.1",

"MA0171.1","MA0172.1","MA0173.1","MA0174.1","MA0175.1","MA0176.1","MA0177.1",
"MA0178.1","MA0179.1","MA0180.1",

"MA0181.1","MA0182.1","MA0183.1","MA0184.1","MA0185.1","MA0186.1","MA0187.1",
"MA0188.1","MA0189.1","MA0190.1",

"MA0191.1","MA0192.1","MA0193.1","MA0194.1","MA0195.1","MA0196.1","MA0197.1",
"MA0198.1","MA0199.1","MA0200.1",

"MA0201.1","MA0202.1","MA0203.1","MA0204.1","MA0205.1","MA0206.1","MA0207.1",
"MA0208.1","MA0209.1","MA0210.1",

"MA0211.1","MA0212.1","MA0213.1","MA0214.1","MA0215.1","MA0216.1","MA0217.1",
"MA0218.1","MA0219.1","MA0220.1",

"MA0221.1","MA0222.1","MA0224.1","MA0225.1","MA0226.1","MA0227.1","MA0228.1",
"MA0229.1","MA0230.1","MA0231.1",

"MA0232.1","MA0233.1","MA0234.1","MA0235.1","MA0236.1","MA0237.1","MA0238.1",
"MA0239.1","MA0240.1","MA0241.1",

"MA0242.1","MA0243.1","MA0244.1","MA0245.1","MA0246.1","MA0247.1","MA0248.1",
"MA0249.1","MA0250.1","MA0251.1",

"MA0252.1","MA0253.1","MA0254.1","MA0255.1","MA0256.1","MA0257.1","MA0094.2",
"MA0443.1","MA0444.1","MA0445.1",

"MA0446.1","MA0447.1","MA0448.1","MA0449.1","MA0450.1","MA0451.1","MA0452.1",
"MA0197.2","MA0454.1","MA0456.1",

"MA0457.1","MA0458.1","MA0459.1","MA0460.1","MA0529.1","MA0216.2","MA0530.1",
"MA0531.1","MA0452.2","MA0532.1",

"MA0533.1","MA0534.1","MA0237.2","MA0535.1","MA0536.1","MA0247.2","MA0915.1",
"MA0916.1","MA0917.1","MA0086.2",

"MA1455.1","MA1456.1","MA1457.1","MA1458.1","MA1459.1","MA1460.1","MA1461.1",
"MA1462.1","MA0529.2","MA0205.2",

"MA0249.2","MA1700.1","MA1702.1","MA1836.1","MA1837.1","MA1838.1","MA1839.1",
"MA1840.1","MA1841.1","MA2107.1",

"MA2108.1","MA2188.1","MA2189.1","MA2190.1","MA2191.1","MA2192.1","MA2193.1",
"MA2194.1","MA2195.1","MA2196.1",

"MA2197.1","MA2198.1","MA2199.1","MA2200.1","MA2201.1","MA2202.1","MA2203.1",
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"MA2247.1","MA2248.1","MA2249.1","MA2250.1","MA2251.1","MA2252.1","MA2253.1",
"MA2254.1","MA2255.1","MA2256.1",

"MA2257.1","MA2258.1","MA2259.1","MA2260.1","MA2261.1","MA2262.1","MA2263.1",
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"MA2267.1","MA2268.1","MA2269.1","MA2270.1","MA2271.1","MA2272.1","MA2273.1",
"MA2274.1","MA2275.1","MA2276.1",

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"MA2284.1","MA2285.1","MA2286.1",

"MA2287.1","MA2288.1","MA2289.1","MA2290.1","MA2291.1","MA2292.1","MA2293.1",
"MA2294.1","MA2463.1","MA2464.1",

"MA2465.1","MA2466.1","MA2467.1","MA2468.1","MA2295.1","MA2296.1","MA2297.1",
"MA2298.1","MA2299.1","MA2300.1",

"MA2301.1","MA2302.1","MA2303.1","MA2304.1","MA2305.1","MA2306.1","MA2307.1",
"MA2308.1","MA2309.1","MA2310.1",

"MA2311.1","MA2312.1","MA2313.1","MA2314.1","MA2315.1","MA2316.1","MA2317.1",
"MA2318.1","MA2319.1","MA2320.1",

"MA2321.1","MA2322.1","MA0529.3","MA0010.2","MA0011.2","MA0012.2","MA0443.2",
"MA0015.2","MA0184.2","MA0530.2",

"MA0531.2","MA0185.2","MA1455.2","MA1456.2","MA1836.2","MA0915.2","MA0189.2",
"MA1837.2","MA1838.2","MA0222.2",

"MA1839.2","MA1840.2","MA1457.2","MA1458.2","MA1841.2","MA0451.2","MA0452.3",
"MA0195.2","MA1459.2","MA0197.3",

"MA0126.2","MA1702.2","MA0457.2","MA1460.2","MA0536.2","MA0200.2","MA0239.2",
"MA0243.2","MA0458.2","MA0086.3",

"MA0532.2","MA1461.2","MA0247.3","MA0205.3","MA0249.3","MA0094.3","MA0016.2",
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"MA0547.1","MA0918.1","MA0919.1","MA0920.1","MA0921.1","MA0922.1","MA0923.1",
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"MA1703.1","MA1704.1","MA2127.1",

"MA2128.1","MA2129.1","MA2130.1","MA2131.1","MA2132.1","MA2133.1","MA2134.1",
"MA2135.1","MA2136.1","MA2137.1",

"MA2138.1","MA2139.1","MA2140.1","MA2141.1","MA2142.1","MA2143.1","MA2144.1",
"MA2145.1","MA2146.1","MA2147.1",

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"MA2155.1","MA2156.1","MA2157.1",

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"MA2168.1","MA2169.1","MA2170.1","MA2171.1","MA2172.1","MA2173.1","MA2174.1",
"MA2175.1","MA2176.1","MA2177.1",

"MA2178.1","MA2179.1","MA2180.1","MA2181.1","MA2182.1","MA2183.1","MA2184.1",
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"MA1438.2","MA1444.2","MA0264.2","MA1699.2","MA0921.2","MA0538.2","MA1446.2",
"MA0919.2","MA0541.2","MA1701.2",

"MA1439.2","MA0543.2","MA0920.2","MA0545.2","MA1449.2","MA0923.2","MA1441.2",
"MA1450.2","MA1703.2","MA0547.2",

"MA0925.2","MA1443.2","MA0918.2","MA0928.2","MA1704.2","MA0921.3","MA0118.1",
"MA1843.1","MA1844.1","MA1845.1",

"MA1846.1","MA1847.1","MA1848.1","MA1849.1","MA1850.1","MA1851.1","MA1852.1",
"MA1853.1","MA1854.1","MA1855.1",

"MA1856.1","MA1857.1","MA1858.1","MA1859.1","MA1860.1","MA1861.1","MA1862.1",
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"MA1866.1","MA1867.1","MA1868.1","MA1869.1","MA1870.1","MA1871.1","MA1872.1",
"MA1873.1","MA1874.1","MA1875.1",

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"MA1883.1","MA1884.1","MA1885.1",

"MA1886.1","MA1887.1","MA1888.1","MA1889.1","MA1890.1","MA1891.1","MA1892.1",
"MA1893.1","MA1894.1","MA1895.1",

"MA1896.1","MA1897.1","MA1898.1","MA1899.1","MA1900.1","MA1901.1","MA1902.1",
"MA1903.1","MA1904.1","MA1905.1",

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```

```
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"MA2330.2","MA1589.3","MA0088.3","MA1976.3","MA0116.2"
```

```
])
```

```
def motif_id_from_header(header: str) -> str:  
    return header.lstrip(">").split()[0]
```

```
with open(jaspar_file, "r", encoding="utf-8") as fh:  
    lines = fh.readlines()
```

```
out = []  
i = 0  
while i < len(lines):  
    line = lines[i]  
    if line.startswith(">"):  
        motif_id = motif_id_from_header(line)  
        block = lines[i:i+5]  
        if motif_id in allowed_motif_ids:  
            out.extend(block)  
        i += 5  
    else:  
        i += 1
```

```
if not out:  
    raise SystemExit("No motifs matched the whitelist.")
```

```
with open(filtered_out, "w", encoding="utf-8") as fh:  
    fh.writelines(out)
```

```
print(f"Wrote filtered motif file with {len(out)//5} motifs to: {filtered_out}")
```

```
PY
```

```
#####
```

```
#####
```

```
# Merging BAMs
```

```
#####
```

```
#####
```

```
samtools merge -@ 8 "$OUT/merged_bam/0h_merged.bam" \
```

```
"$BAMDIR/0hR1.dedup.bam" "$BAMDIR/0hR2.dedup.bam"
```

```
"$BAMDIR/0hR3.dedup.bam"
```

```
samtools sort -@ 8 -o "$OUT/merged_bam/0h_merged.sorted.bam"
```

```
"$OUT/merged_bam/0h_merged.bam"
```

```
samtools index "$OUT/merged_bam/0h_merged.sorted.bam"
```

```
samtools merge -@ 8 "$OUT/merged_bam/3h_merged.bam" \
"$BAMDIR/3hR1.dedup.bam" "$BAMDIR/3hR2.dedup.bam"
"$BAMDIR/3hR3.dedup.bam"
samtools sort -@ 8 -o "$OUT/merged_bam/3h_merged.sorted.bam"
"$OUT/merged_bam/3h_merged.bam"
samtools index "$OUT/merged_bam/3h_merged.sorted.bam"
```

```
samtools merge -@ 8 "$OUT/merged_bam/12h_merged.bam" \
"$BAMDIR/12hR1.dedup.bam" "$BAMDIR/12hR2.dedup.bam"
"$BAMDIR/12hR3.dedup.bam"
samtools sort -@ 8 -o "$OUT/merged_bam/12h_merged.sorted.bam"
"$OUT/merged_bam/12h_merged.bam"
samtools index "$OUT/merged_bam/12h_merged.sorted.bam"
```

```
samtools merge -@ 8 "$OUT/merged_bam/24h_merged.bam" \
"$BAMDIR/24hR1.dedup.bam" "$BAMDIR/24hR2.dedup.bam"
"$BAMDIR/24hR3.dedup.bam"
samtools sort -@ 8 -o "$OUT/merged_bam/24h_merged.sorted.bam"
"$OUT/merged_bam/24h_merged.bam"
samtools index "$OUT/merged_bam/24h_merged.sorted.bam"
```

```
echo "All BAMs merged, sorted, and indexed"
```

```
#####
#####
```

```
# 3) TOBIAS ATACorrect
```

```
#####
#####
```

```
conda activate Tobias_env
```

```
cd "$TOBIAS_DIR"
```

```
TOBIAS ATACorrect \
--bam "$OUT/merged_bam/0h_merged.sorted.bam" \
--genome "$GENOME" \
--peaks "$PEAKS" \
--outdir "$OUT/ATACorrect/0h" \
> "$OUT/logs/0h_ATACorrect.log" 2>&1
```

```
TOBIAS ATACorrect \
--bam "$OUT/merged_bam/3h_merged.sorted.bam" \
--genome "$GENOME" \
--peaks "$PEAKS" \
--outdir "$OUT/ATACorrect/3h" \
> "$OUT/logs/3h_ATACorrect.log" 2>&1
```

```
TOBIAS ATACorrect \
--bam "$OUT/merged_bam/12h_merged.sorted.bam" \
```

```
--genome "$GENOME" \
--peaks "$PEAKS" \
--outdir "$OUT/ATACorrect/12h" \
> "$OUT/logs/12h_ATACorrect.log" 2>&1
```

```
TOBIAS ATACorrect \
--bam "$OUT/merged_bam/24h_merged.sorted.bam" \
--genome "$GENOME" \
--peaks "$PEAKS" \
--outdir "$OUT/ATACorrect/24h" \
> "$OUT/logs/24h_ATACorrect.log" 2>&1
```

```
echo "ATACorrect finished"
```

```
#####
#####
```

```
# 4) TOBIAS ScoreBigwig
```

```
#####
#####
```

```
TOBIAS ScoreBigwig \
--signal "$OUT/ATACorrect/0h/0h_merged.sorted_corrected.bw" \
--regions "$PEAKS" \
--output "$OUT/ScoreBigwig/0h_footprints.bw"
```

```
TOBIAS ScoreBigwig \
--signal "$OUT/ATACorrect/3h/3h_merged.sorted_corrected.bw" \
--regions "$PEAKS" \
--output "$OUT/ScoreBigwig/3h_footprints.bw"
```

```
TOBIAS ScoreBigwig \
--signal "$OUT/ATACorrect/12h/12h_merged.sorted_corrected.bw" \
--regions "$PEAKS" \
--output "$OUT/ScoreBigwig/12h_footprints.bw"
```

```
TOBIAS ScoreBigwig \
--signal "$OUT/ATACorrect/24h/24h_merged.sorted_corrected.bw" \
--regions "$PEAKS" \
--output "$OUT/ScoreBigwig/24h_footprints.bw"
```

```
echo "ScoreBigwig finished"
```

```
#####
#####
```

```
# 5) TOBIAS BINDetect using only the metazoan motifs
```

```
#####
#####
```

```
conda activate Tobias_env_bindetect
```

```
ulimit -n 8192
```

```
TOBIAS BINDetect \  
--cores 1 \  
--motifs "$FILTERED_MOTIFS" \  
--signals \  
"$OUT/ScoreBigwig/0h_footprints.bw" \  
"$OUT/ScoreBigwig/3h_footprints.bw" \  
"$OUT/ScoreBigwig/12h_footprints.bw" \  
"$OUT/ScoreBigwig/24h_footprints.bw" \  
--genome "$GENOME" \  
--peaks "$PEAKS" \  
--cond_names 0h 3h 12h 24h \  
--outdir "$OUT/BINDetect" \  
> "$OUT/BINDetect/bindetect.log" 2>&1
```

```
echo "BINDetect finished"
```

```
chmod +x run_tobias_whitelist.sh
```

```
bash -n run_tobias_whitelist.sh
```

```
./run_tobias_whitelist.sh
```

```
ls -lh /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis/TOBIAS/TOBIAS_hypoxia/hypoxia_ATAC_peaks.animal_onl  
y_whitelist.jaspar
```

```
tail -n 50 /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis/TOBIAS/TOBIAS_hypoxia/logs/0h_ATACCorrect.log
```

```
ls -lh /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis/TOBIAS/TOBIAS_hypoxia/ATACCorrect/0h  
ls -lh /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis/TOBIAS/TOBIAS_hypoxia/ScoreBigwig  
ls -lh /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis/TOBIAS/TOBIAS_hypoxia/BINDetect
```

```
tail -n 50 /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis/TOBIAS/TOBIAS_hypoxia/BINDetect/bindetect.log
```

Constructing genome track figures

```
cd /drives/raid/AboobakerLab/James/hypoxia_ATAC-seq_time-  
course/april_2026_analysis/TOBIAS/TOBIAS_hypoxia
```

```
mkdir -p ATAC_bigwigs
```

```
conda install -c bioconda deeptools
```

```
bamCoverage -b merged_bam/0h_merged.sorted.bam -o ATAC_bigwigs/0h_ATAC.bw --  
normalizeUsing CPM --binSize 10 -p 8 --verbose
```

```
bamCoverage -b merged_bam/3h_merged.sorted.bam -o ATAC_bigwigs/3h_ATAC.bw --  
normalizeUsing CPM --binSize 10 -p 8 --verbose
```

```
bamCoverage -b merged_bam/12h_merged.sorted.bam -o  
ATAC_bigwigs/12h_ATAC.bw --normalizeUsing CPM --binSize 10 -p 8 --verbose
```

```
bamCoverage -b merged_bam/24h_merged.sorted.bam -o  
ATAC_bigwigs/24h_ATAC.bw --normalizeUsing CPM --binSize 10 -p 8 --verbose
```

```
grep "^#" ../../Smed_Rink.gtf > ../../Smed_Rink.sorted.gtf
```

```
grep -v "^#" ../../Smed_Rink.gtf | sort -k1,1 -k4,4n >> ../../Smed_Rink.sorted.gtf
```